Experiment No: 05

Experiment Name: Verification of Super Position theorem.

Course Code: EEE-1209

Course Title: Electrical Circuit lab

Submitted to:

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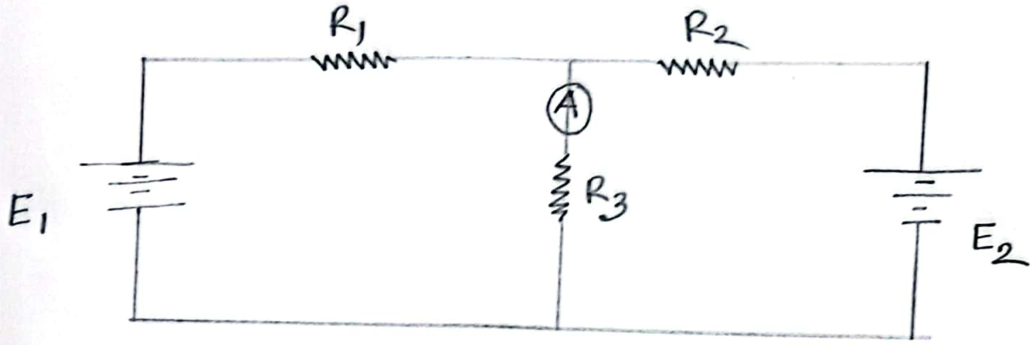
Experiment name: verification of superposition theorem

Superposition Theorem: Superposition theorem states that in any linear, active, bilateral network having more than one source, the response across any element is the sum of the responses obtained from each source considered separately and all other sources are replaced by their internal resistance. The superposition theorem is used to solve the network where two or more sources are present and connected.

Required eqipments:

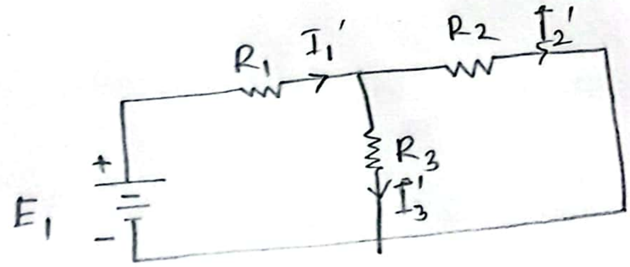
1. Medium transmission line 'T' model
2. AC power supply
3. AC Voltagemeter
4. AC Ammeter
5. Multimeter
6. single phase Autotransformer
7. Connecting wires

Circuit construction :



Result & Discussion :

First take the source E_1 alone and short circuit the E_2 source as shown in the circuit diagram below



Here, the value of current flowing in each branch, i.e. I_1', I_2' and I_3' is calculated by the following equations.

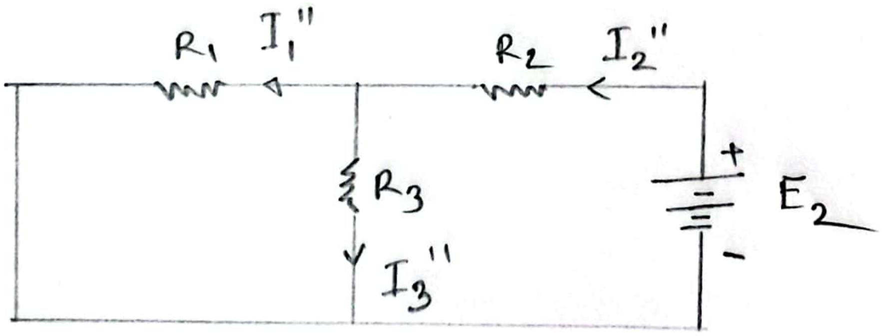
$$I_1' = \frac{E_1}{\frac{R_2 R_3}{R_2 + R_3} + R_1} \quad \cdots \quad (1)$$

$$I_2' = I_1' \frac{R_3}{R_2+R_3} \quad \text{---} (11)$$

The difference between the above two equations gives the value of the current I_3'.

$$I_3' = I_1' - I_2'$$

Now, activating the voltage source V, and deactivating the voltage source V by short circuiting it, find the various currents i.e. I_1'', I_2'' and I_3'' flowing in the circuit diagram shown below



Here,
$$I_2'' = \frac{V_2}{\frac{R_1 R_3}{R_1+R_3} + R_2}$$

$$I_1'' = \frac{R_3}{R_1+R_3} I_2''$$

And the value of the current I_3'' will be calculated by the equation shown below

$$I_3'' = I_2'' - I_1''$$

As per the superposition theorem the value of current I_1, I_2, I_3 is now calculated as

$$I_3 = I_3' - I_3''$$

$$I_2 = I_2' - I_2''$$

$$I_1 = I_1' - I_1''$$

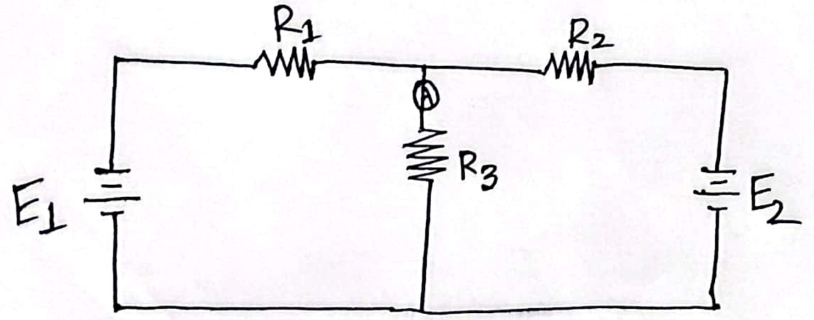
Direction of current should be taken care while finding the current in the various branches

Data table:

E_1	E_2	I when E_1 & E_2 both active	I_1	I_1	$I_{exp} = I_1 + I_2$	Error
5V	10V	1.4 mA	0.4mA	0.9mA	1.3 mA	7.14 %
10V	15V	2.4 mA	0.9mA	1.4mA	2.3 mA	4.1 %

Conclusion: The aim of this experiment was to verify the superposition theorem. We have determined that the current through any resistor branch is the algebraic sum of the contributions due to each source active independently.

xperiment name: Verification of superposition theorem



$R_1 = R_2 = 0.98 k\Omega$

$R_3 = 4.65 k\Omega$

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E_1	E_2	I, when E_1 & E_2 both active	I_1	I_2	$I_{exp} = I_1 + I_2$	Error
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